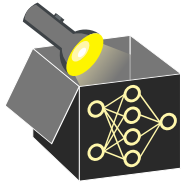
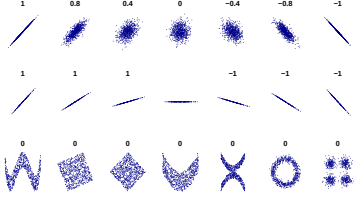
# Interpretable Machine Learning



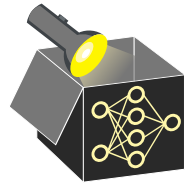
## Intro to IML
## Correlation and Dependencies




**Learning goals**

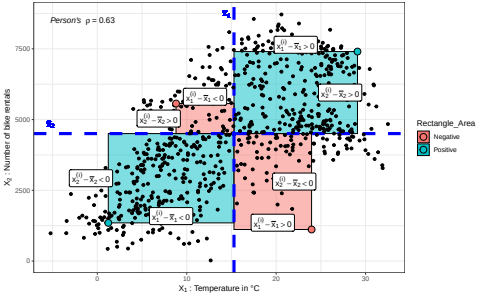
- Pearson correlation
- Coefficient of determination $R^2$
- Mutual information
- Correlation vs. dependence

# PEARSON'S CORRELATION COEFFICIENT $\rho$



**Correlation** often refers to Pearson's correlation (captures only **linear relationship**)

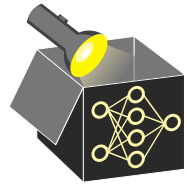
$$\rho(X_1, X_2) = \frac{\sum_{i=1}^{n}(x_1^{(i)} - \bar{x}_1)(x_2^{(i)} - \bar{x}_2)}{\sqrt{\sum_{i=1}^{n}(x_1^{(i)} - \bar{x}_1)^2 \sum_{i=1}^{n}(x_2^{(i)} - \bar{x}_2)^2}} \in [-1, 1]$$




Geometric interpretation of $\rho$:

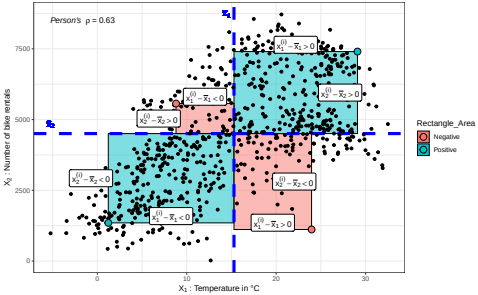
- Numerator is sum of rectangle's area with width $x_1^{(i)} - \bar{x}_1$ and height $x_2^{(i)} - \bar{x}_2$
- Areas enter numerator with positive (**+**) or negative (**-**) sign, depending on position
- Denominator scales the sum into the range $[-1, 1]$

# PEARSON'S CORRELATION COEFFICIENT $\rho$



**Correlation** often refers to Pearson's correlation (captures only **linear relationship**)

$$\rho(X_1, X_2) = \frac{\sum_{i=1}^{n}(x_1^{(i)} - \bar{x}_1)(x_2^{(i)} - \bar{x}_2)}{\sqrt{\sum_{i=1}^{n}(x_1^{(i)} - \bar{x}_1)^2 \sum_{i=1}^{n}(x_2^{(i)} - \bar{x}_2)^2}} \in [-1, 1]$$



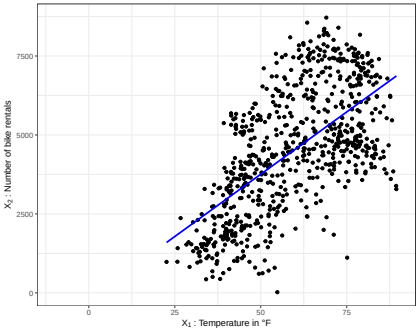

Geometric interpretation of $\rho$:

- Numerator is sum of rectangle's area with width $x_1^{(i)} - \bar{x}_1$ and height $x_2^{(i)} - \bar{x}_2$
- Areas enter numerator with positive (**+**) or negative (**-**) sign, depending on position
- Denominator scales the sum into the range $[-1, 1]$

- $\rho > 0$ if positive areas dominate negative areas $\rightsquigarrow$ $X_1$, $X_2$ positive correlated
- $\rho < 0$ if negative areas dominate positive areas $\rightsquigarrow$ $X_1$, $X_2$ negative correlated
- $\rho = 0$ if area of rectangles cancels out$\rightsquigarrow$ $X_1$, $X_2$ linearly uncorrelated

# COEFFICIENT OF DETERMINATION $R^2$

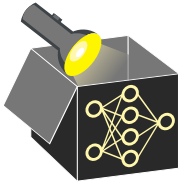
Another method to evaluate **linear dependency** between features is $R^2$



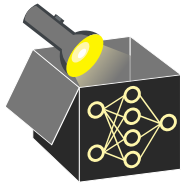

- Fit a linear model:
  $\hat{x}_2 = \hat{f}_{LM}(x_1) = \theta_0 + \theta_1 x_1$

$\rightsquigarrow$ Slope $\theta_1 = 0 \Rightarrow$ no dependence

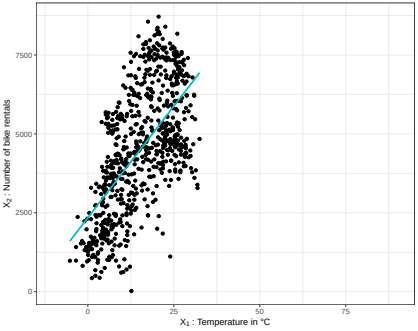
$\rightsquigarrow$ Large slope $\Rightarrow$ strong dependence



# COEFFICIENT OF DETERMINATION $R^2$



Another method to evaluate **linear dependency** between features is $R^2$




- Fit a linear model:
  $\hat{x}_2 = \hat{f}_{LM}(x_1) = \theta_0 + \theta_1 x_1$

$\rightsquigarrow$ Slope $\theta_1 = 0 \Rightarrow$ no dependence

$\rightsquigarrow$ Large slope $\Rightarrow$ strong dependence

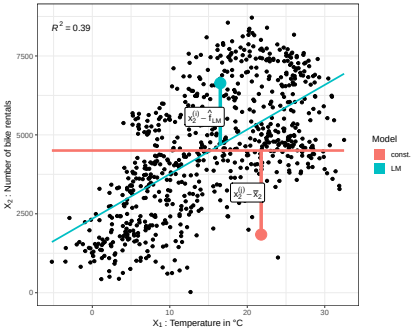
- Exact $\theta_1$ score problematic

$\rightsquigarrow$ Re-scaling of $x_1$ or $x_2$ changes $\theta_1$

$\rightsquigarrow$ °F $\rightarrow$ °C $\Rightarrow \theta_1 = 78 \rightarrow \theta_1^* = 141$

# COEFFICIENT OF DETERMINATION $R^2$

Another method to evaluate **linear dependency** between features is $R^2$




- Fit a linear model:
  $\hat{x}_2 = \hat{f}_{LM}(x_1) = \theta_0 + \theta_1 x_1$

$\rightsquigarrow$ Slope $\theta_1 = 0 \Rightarrow$ no dependence

$\rightsquigarrow$ Large slope $\Rightarrow$ strong dependence

- Exact $\theta_1$ score problematic

$\rightsquigarrow$ Re-scaling of $x_1$ or $x_2$ changes $\theta_1$

- Set $SSE_{LM}$ in relation to $SSE$ of a constant model $\hat{f}_c = \bar{x}_2$

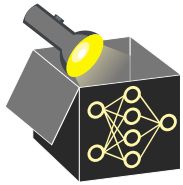
$$SSE_{LM} = \sum_{i=1}^{n} (x_2^{(i)} - \hat{f}_{LM}(x_1^{(i)}))^2$$

$$SSE_c = \sum_{i=1}^{n} (x_2^{(i)} - \bar{x}_2)^2$$

$\Rightarrow$ Measure of fitting quality of LM: $R^2 = 1 - \frac{SSE_{LM}}{SSE_c} \in [0, 1]$

$\Rightarrow \rho(X_1, X_2) = R$

# JOINT, MARGINAL AND CONDITIONAL DISTRIBUTION



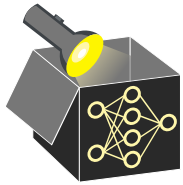
For two discrete random variables $X_1$, $X_2$:

**Joint distribution**

$$p_{X_1,X_2}(x_1, x_2) = \mathbb{P}(X_1 = x_1, X_2 = x_2)$$

| $p_{X_1,X_2}$ | $\mathbb{P}(X_2 = 0)$ | $\mathbb{P}(X_2 = 1)$ | $p_{X_1}$ |
|---|---|---|---|
| $\mathbb{P}(X_1 = 0)$ | 0.2 | 0.3 | 0.5 |
| $\mathbb{P}(X_1 = 1)$ | 0.1 | 0.4 | 0.5 |
| $p_{X_2}$ | 0.3 | 0.7 | 1 |

# JOINT, MARGINAL AND CONDITIONAL DISTRIBUTION



For two discrete random variables $X_1$, $X_2$:

**Joint distribution**

$$p_{X_1,X_2}(x_1, x_2) = \mathbb{P}(X_1 = x_1, X_2 = x_2)$$

| $p_{X_1,X_2}$ | $\mathbb{P}(X_2 = 0)$ | $\mathbb{P}(X_2 = 1)$ | $p_{X_1}$ |
|---|---|---|---|
| $\mathbb{P}(X_1 = 0)$ | 0.2 | 0.3 | 0.5 |
| $\mathbb{P}(X_1 = 1)$ | 0.1 | 0.4 | 0.5 |
| $p_{X_2}$ | 0.3 | 0.7 | 1 |

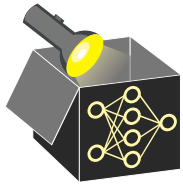
**Marginal distribution**

$$p_{X_1}(x_1) = \mathbb{P}(X_1 = x_1) = \sum_{x_2 \in \mathcal{X}_2} p(x_1, x_2)$$

| $p_{X_1,X_2}$ | $\mathbb{P}(X_2 = 0)$ | $\mathbb{P}(X_2 = 1)$ | $p_{X_1}$ |
|---|---|---|---|
| $\mathbb{P}(X_1 = 0)$ | 0.2 | 0.3 | 0.5 |
| $\mathbb{P}(X_1 = 1)$ | 0.1 | 0.4 | 0.5 |
| $p_{X_2}$ | 0.3 | 0.7 | 1 |

$\rightsquigarrow$ In continuous case with integrals

# JOINT, MARGINAL AND CONDITIONAL DISTRIBUTION



For two discrete random variables $X_1$, $X_2$:

**Joint distribution**

$$p_{X_1,X_2}(x_1, x_2) = \mathbb{P}(X_1 = x_1, X_2 = x_2)$$

| $p_{X_1,X_2}$ | $\mathbb{P}(X_2 = 0)$ | $\mathbb{P}(X_2 = 1)$ | $p_{X_1}$ |
|---|---|---|---|
| $\mathbb{P}(X_1 = 0)$ | 0.2 | 0.3 | 0.5 |
| $\mathbb{P}(X_1 = 1)$ | 0.1 | 0.4 | 0.5 |
| $p_{X_2}$ | 0.3 | 0.7 | 1 |

**Marginal distribution**

$$p_{X_1}(x_1) = \mathbb{P}(X_1 = x_1) = \sum_{x_2 \in \mathcal{X}_2} p(x_1, x_2)$$

| $p_{X_1,X_2}$ | $\mathbb{P}(X_2 = 0)$ | $\mathbb{P}(X_2 = 1)$ | $p_{X_1}$ |
|---|---|---|---|
| $\mathbb{P}(X_1 = 0)$ | 0.2 | 0.3 | 0.5 |
| $\mathbb{P}(X_1 = 1)$ | 0.1 | 0.4 | 0.5 |
| $p_{X_2}$ | 0.3 | 0.7 | 1 |

$\rightsquigarrow$ In continuous case with integrals

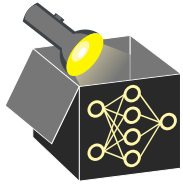
**Conditional distribution**

$$p_{X_1|X_2}(x_1 \mid x_2) = \mathbb{P}(X_1 = x_1 \mid X_2 = x_2) = \frac{p_{X_1,X_2}(x_1, x_2)}{p_{X_2}(x_2)}$$

| | $x_2 = 0$ | $x_2 = 1$ |
|---|---|---|
| $\mathbb{P}(X_1 = 0 \mid X_2 = x_2)$ | 0.67 | 0.43 |
| $\mathbb{P}(X_1 = 1 \mid X_2 = x_2)$ | 0.33 | 0.57 |
| $\sum$ | 1 | 1 |

# DEPENDENCE

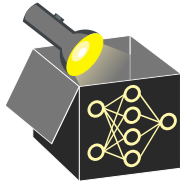
**Dependence:** Describes general dependence structure (e.g., non-lin. relationships)



- Definition: $X_j$, $X_k$ independent $\Leftrightarrow$ joint distribution is product of marginals:

$$\mathbb{P}(X_j, X_k) = \mathbb{P}(X_j) \cdot \mathbb{P}(X_k)$$

# DEPENDENCE



**Dependence:** Describes general dependence structure (e.g., non-lin. relationships)

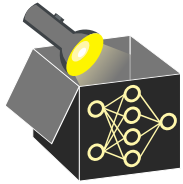
- Definition: $X_j$, $X_k$ independent $\Leftrightarrow$ joint distribution is product of marginals:

$$\mathbb{P}(X_j, X_k) = \mathbb{P}(X_j) \cdot \mathbb{P}(X_k)$$

- Equivalent definition (knowing $X_k$ gives no info about $X_j$ and vice versa):

$$\mathbb{P}(X_j|X_k) = \mathbb{P}(X_j) \text{ and } \mathbb{P}(X_k|X_j) = \mathbb{P}(X_k) \text{ (follows from cond. probability)}$$

# DEPENDENCE



**Dependence:** Describes general dependence structure (e.g., non-lin. relationships)

- Definition: $X_j$, $X_k$ independent $\Leftrightarrow$ joint distribution is product of marginals:

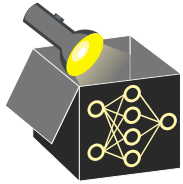
$$\mathbb{P}(X_j, X_k) = \mathbb{P}(X_j) \cdot \mathbb{P}(X_k)$$

- Equivalent definition (knowing $X_k$ gives no info about $X_j$ and vice versa):

$$\mathbb{P}(X_j|X_k) = \mathbb{P}(X_j) \text{ and } \mathbb{P}(X_k|X_j) = \mathbb{P}(X_k) \text{ (follows from cond. probability)}$$

- Measuring complex dependencies is difficult but different measures exist
  Examples
  $\rightsquigarrow$ Spearman correlation (measures monotonic dependencies via ranks)
  $\rightsquigarrow$ Information-theoretical measures like mutual information
  $\rightsquigarrow$ Kernel-based measures like Hilbert-Schmidt Independence Criterion (HSIC)

# DEPENDENCE



**Dependence:** Describes general dependence structure (e.g., non-lin. relationships)

- Definition: $X_j$, $X_k$ independent $\Leftrightarrow$ joint distribution is product of marginals:

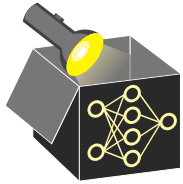
$$\mathbb{P}(X_j, X_k) = \mathbb{P}(X_j) \cdot \mathbb{P}(X_k)$$

- Equivalent definition (knowing $X_k$ gives no info about $X_j$ and vice versa):

$$\mathbb{P}(X_j|X_k) = \mathbb{P}(X_j) \text{ and } \mathbb{P}(X_k|X_j) = \mathbb{P}(X_k) \text{ (follows from cond. probability)}$$

- Measuring complex dependencies is difficult but different measures exist
  Examples
  $\rightsquigarrow$ Spearman correlation (measures monotonic dependencies via ranks)
  $\rightsquigarrow$ Information-theoretical measures like mutual information
  $\rightsquigarrow$ Kernel-based measures like Hilbert-Schmidt Independence Criterion (HSIC)
- **N.B.:** $X_j$, $X_k$ indep. $\Rightarrow \rho(X_j, X_k) = 0$ **but** $\rho(X_j, X_k) = 0 \not\Rightarrow X_j, X_k$ indep.
  Equivalency holds if distribution is jointly normal

# MUTUAL INFORMATION

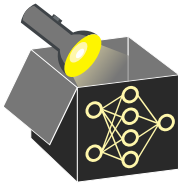


- MI describes expected amount of information shared by two RVs:

$$MI(X_1, X_2) = \mathbb{E}_{p(x_1,x_2)}\left[log\left(\frac{p(x_1,x_2)}{p(x_1)p(x_2)}\right)\right]$$

- MI measures amount of "dependence" between features by looking how different the joint distribution is from pure indep. $p(x_1, x_2) = p(x_1)p(x_2)$
  $\rightsquigarrow MI(X_1, X_2) = \mathbb{E}_{p(x_1,x_2)}\left[log\left(\frac{p(x_1,x_2)}{p(x_1,x_2)}\right)\right] = \mathbb{E}_{p(x_1,x_2)}\left[log(1)\right] = 0$
  $\rightsquigarrow MI(X_j, X_k) = 0$ if and only if the features are independent
- Unlike (Pearson) correlation, MI is also defined for categorical features

# MUTUAL INFORMATION: EXAMPLE



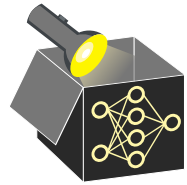
For two discrete RV $X_1$ and $Y$:

$$MI(X_1; Y) = \mathbb{E}_{p(x_1,y)}\left[log\left(\frac{p(x_1,y)}{p(x_1)p(y)}\right)\right] = \sum_{x_1 \in \mathcal{X}_1}\sum_{y \in \mathcal{Y}} p(x_1,y) log\left(\frac{p(x_1,y)}{p(x_1)p(y)}\right)$$

| $\mathbf{X}_1$ | ... | $\mathbf{Y}$ |
|---|---|---|
| yes | ... | yes |
| yes | ... | no |
| no | ... | yes |
| no | ... | no |

| | $\mathbb{P}(X_1 = \text{yes})$ | $\mathbb{P}(X_1 = \text{no})$ | $p_Y$ |
|---|---|---|---|
| $\mathbb{P}(Y = \text{yes})$ | 0.25 | 0.25 | 0.5 |
| $\mathbb{P}(Y = \text{no})$ | 0.25 | 0.25 | 0.5 |
| $p_{X_1}$ | 0.5 | 0.5 | 1 |

# MUTUAL INFORMATION: EXAMPLE



For two discrete RV $X_1$ and $Y$:

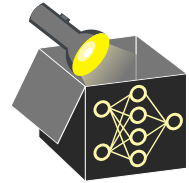
$$MI(X_1; Y) = \mathbb{E}_{p(x_1,y)}\left[log\left(\frac{p(x_1,y)}{p(x_1)p(y)}\right)\right] = \sum_{x_1 \in \mathcal{X}_1}\sum_{y \in \mathcal{Y}} p(x_1,y) log\left(\frac{p(x_1,y)}{p(x_1)p(y)}\right)$$

| $X_1$ | ... | Y |
|---|---|---|
| yes | ... | yes |
| yes | ... | no |
| no | ... | yes |
| no | ... | no |

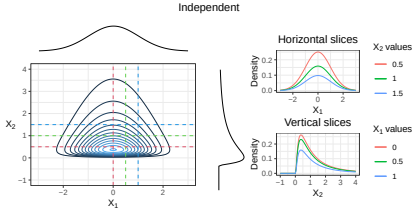
| | $\mathbb{P}(X_1 = \text{yes})$ | $\mathbb{P}(X_1 = \text{no})$ | $p_Y$ |
|---|---|---|---|
| $\mathbb{P}(Y = \text{yes})$ | 0.25 | 0.25 | 0.5 |
| $\mathbb{P}(Y = \text{no})$ | 0.25 | 0.25 | 0.5 |
| $p_{X_1}$ | 0.5 | 0.5 | 1 |

$$\begin{aligned}
MI(X_1; Y) &= 0.25\, log\left(\frac{0.25}{0.5 \cdot 0.5}\right) + 0.25\, log\left(\frac{0.25}{0.5 \cdot 0.5}\right) \\
&\quad + 0.25\, log\left(\frac{0.25}{0.5 \cdot 0.5}\right) + 0.25\, log\left(\frac{0.25}{0.5 \cdot 0.5}\right) \\
&= 0.25\, log\left(\frac{0.25}{0.25}\right) \cdot 4 \\
&= 0.25\, log\,(1) \cdot 4 = 0
\end{aligned}$$

# DEPENDENCE AND INDEPENDENCE



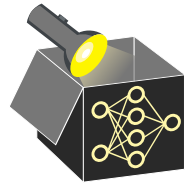
**Example:**



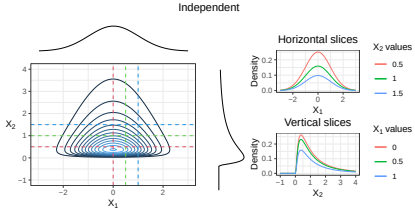

Conditional distributions at different vertical and horizontal slices (after normalizing area to 1) match their marginal distributions

$$\Rightarrow \mathbb{P}(X_1|X_2) = \mathbb{P}(X_1)$$
$$\mathbb{P}(X_2|X_1) = \mathbb{P}(X_2)$$

# DEPENDENCE AND INDEPENDENCE

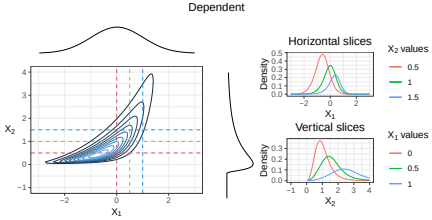


**Example:**




Conditional distributions at different vertical and horizontal slices (after normalizing area to 1) match their marginal distributions

$$\Rightarrow \mathbb{P}(X_1|X_2) = \mathbb{P}(X_1)$$
$$\mathbb{P}(X_2|X_1) = \mathbb{P}(X_2)$$




Conditional distributions do not match their marginal distributions

# CORRELATION VS. DEPENDENCE

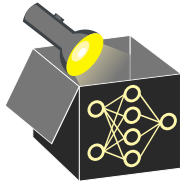
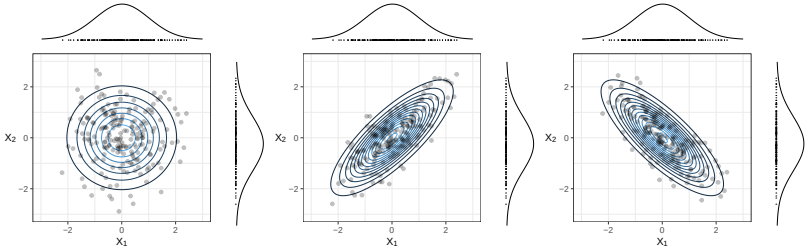


Illustration of bivariate normal distribution with different correlations
$X_1, X_2 \sim N(0, 1)$

$\rho(X_1, X_2) = 0$
(independent)

$\rho(X_1, X_2) = 0.8$

$\rho(X_1, X_2) = -0.8$



# CORRELATION VS. DEPENDENCE

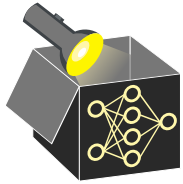
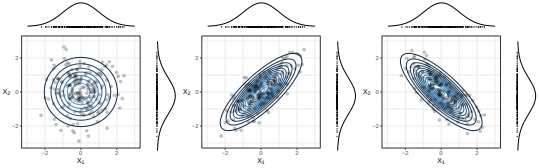


Illustration of bivariate normal distribution with different correlations
$X_1, X_2 \sim N(0, 1)$

$\rho(X_1, X_2) = 0$ (independent)

$\rho(X_1, X_2) = 0.8$

$\rho(X_1, X_2) = -0.8$



Examples with Pearson's corr. $\rho \approx 0$ but non-linear dependencies (MI $\neq 0$):

$\rho(X_1, X_2) = 0$ , $MI(X_1, X_2) = 0.52$ $\rho(X_1, X_2) = 0.01$ , $MI(X_1, X_2) = 0.37$ $\rho(X_1, X_2) = -0.06$ , $MI(X_1, X_2) = 0.61$

